

Imformer-squared

Imformer of imformer implicit modeling of attention

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Introduction

Given a token string $t_1, \dots, t_m$ an Imformer [3] uses a Noken implicit model [1] to jointly learn [2] queries q , keys k , values v and query updates q' . Recall we defined probabilities $\alpha_{ij} = \text{softmax score}(q'_i, k_j)$, where score is a similarity function, and for token t_i , its contextual representation is defined by $h_i = \text{FFN}_\theta[\sum_j \alpha_{ij} v_j]$, where FFN_θ is a feed forward network with parameters θ , applied element-wise. Then as with transformer attention, one learns θ to minimize perplexity, and then continue with layers of updates.

Now instead of using multiple layers of FFN to update, one could apply a single Noken implicit model once, to the string of contextual representations $h_1, \dots, h_m$ learning refined embedding matrices, and then finish with a FFN_θ layer to minimize perplexity. The embeddings derived from contextual representations are designed to capture and align semantics from context. So in Imformer-squared, instead of using multiple layers of feedforward networks to update queries, apply a single Noken implicit model to contextual representations. The idea is that the (second) Noken implicit model acts like a deep stack of FFN for attention updating. To summarize, Imformer-squared is the concatenation of two implicit models, the first Noken implicit model on tokens learns contextual representations, then on these a second Noken implicit model jointly learns query-key-value embeddings, to capture and align meaning from context. Finally, to minimize perplexity, we apply a feed forward network.

ImFormer²

Implicit model for contextual representations

For token string $\{t_1, \dots, t_m\} \subset \mathbb{R}^r$

learn matrices K, Q, V, Q'

and fixed points $s_i \in \mathbb{R}^s$, $s > r$, such that:

$$s_i = \varphi(Ks_i + Qt_i)$$

$$\varepsilon = Vs_i + Q't_i$$

where $\varepsilon \in \mathbb{R}^s$ and K is well-posed for nonlinear φ .

Let $k_i = Ks_i$, $q_i = Qt_i$, $v_i = Vs_i$ and $q'_i = Q't_i$

Define probabilities $\alpha_{ij} = \text{softmax}_{j \in [m]} \text{score}(q'_i, k_j)$, $i \in [m]$

where score is a similarity function.

For token t_i , define its contextual representation

$$\tilde{t}_i = \sum_{j=1}^m \alpha_{ij} v_j \in \mathbb{R}^s, \quad i \in [m]$$

Implicit model for query-key-value embeddings

For contextual representation string $\{\tilde{t}_1, \dots, \tilde{t}_m\} \subset \mathbb{R}^s$

learn matrices $\tilde{K}, \tilde{Q}, \tilde{V}, \tilde{Q}'$ and $\tilde{s}_i \in \mathbb{R}^t$, $t > s$,

such that: $\tilde{s}_i = \tilde{\varphi}(\tilde{K}\tilde{s}_i + \tilde{Q}\tilde{t}_i)$

$$\tilde{\varepsilon} = \tilde{V}\tilde{s}_i + \tilde{Q}'\tilde{t}_i$$

where $\tilde{\varepsilon} \in \mathbb{R}^t$ and $\tilde{K}$ is well-posed for nonlinear $\tilde{\varphi}$.

As before, define probabilities $\tilde{\alpha}_{ij} = \text{softmax}_{j \in [m]} \text{score}(\tilde{q}'_i, \tilde{k}_j)$, $i \in [m]$

Define update $h_i = \text{FFN}_{\Theta} \left[\sum_{j=1}^m \tilde{\alpha}_{ij} \tilde{v}_j \right]$ of $\tilde{q}'_i$,

where FFN_{Θ} is a Feed Forward network with parameters Θ

applied element-wise to the ~~$\tilde{q}$~~ -vector.

Define probabilities $\beta_{ij} = \text{softmax}_{j \in [m]} \text{score}(h_i, \tilde{k}_j)$, $i \in [m]$

and let $j_i^{\beta} = \arg \max_{j \in [m]} \beta_{ij}$.

Given input token t_i , output $o_{i+1} \triangleq t_{j_i^{\beta}}$, and

learn Θ to minimize perplexity $(o_2, \dots, o_{m+1})$.

□

References

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